Deep Learning for Lung Cancer Detection

Achieving 95.7% Accuracy with VGG16 Transfer Learning

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The Diagnostic Bottleneck

Current Challenges
Manual analysis of CT scans is time-intensive.

Current Challenges
High stakes: Misdiagnosis in Malignant cases is a life-critical error.

Our Objective
Develop a robust AI tool for Benign, Malignant, and Normal classification.

The IQ-OTH/NCCD Dataset

Benign

Malignant

Normal

Source: https://www.kaggle.com/datasets/hamdallak/the-iqothnccd-lung-cancer-dataset

Ensuring Scientific Integrity

The “Honesty” Moment:

Initial tests showed 100% accuracy

Trarning accurent!.
(Identified as Data Leakage).

Action Taken:
Strictly removed duplicates to ensure the model generalizes to new patients.

Initial Custom CNN Performance

The Baseline Reality

The Result:
After cleaning the data, a basic CNN only hit **~45%** accuracy.

Conclusion:
Simple architectures cannot "see" the subtle textures required for medical diagnosis. We needed a "**Specialist**."

Adopting Transfer Learning (VGG16)

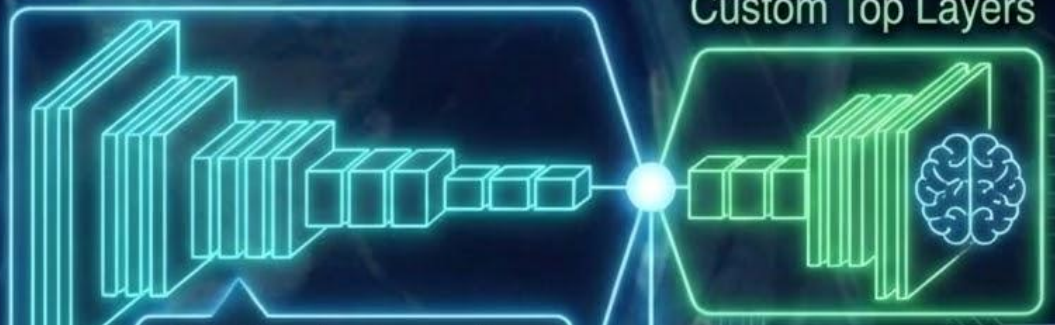
Why VGG16?

Pre-trained on 1.2M images (ImageNet).

VGG16 Model

"Borrow" its eyes:
Fixed Feature Extractor

Retrain only the brain:
Custom Top Layers



Already understands edges, shadows, and complex biological textures.

We "borrow" its eyes and retrain only the brain (top layers).

Building the Medical Expert

Model Architecture

Base: VGG16 (Frozen)

Pre-trained Feature Extractor

Global Average Pooling: Reduces dimensionality.

Takes complex features and simplifies them.

Dropout (0.5): Prevents the model from "cheating" or overfitting.

Randomly drops 50% of neurons during training.

Output: 3-class Softmax

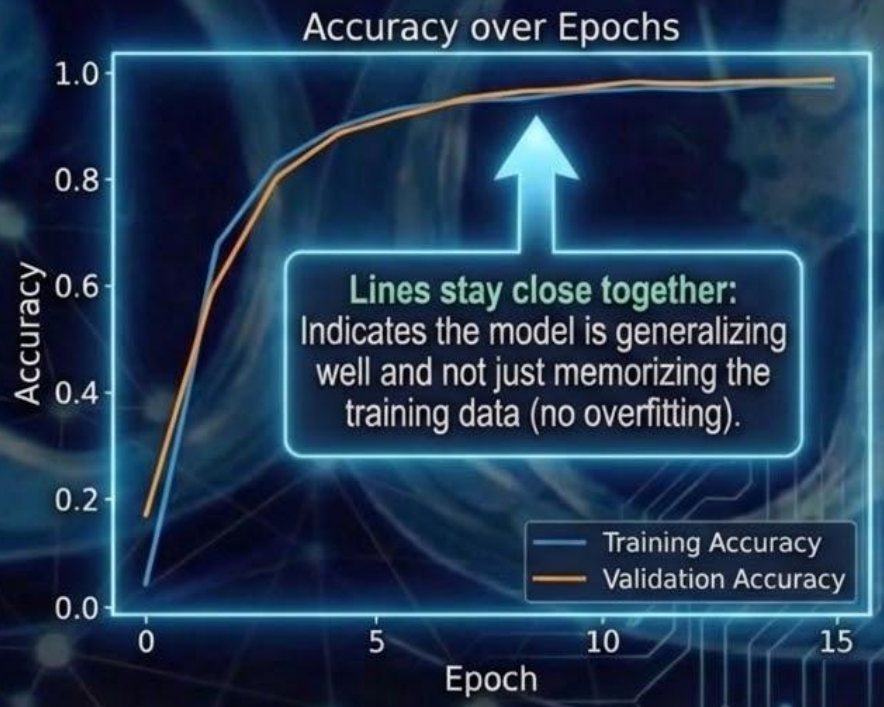
Benign

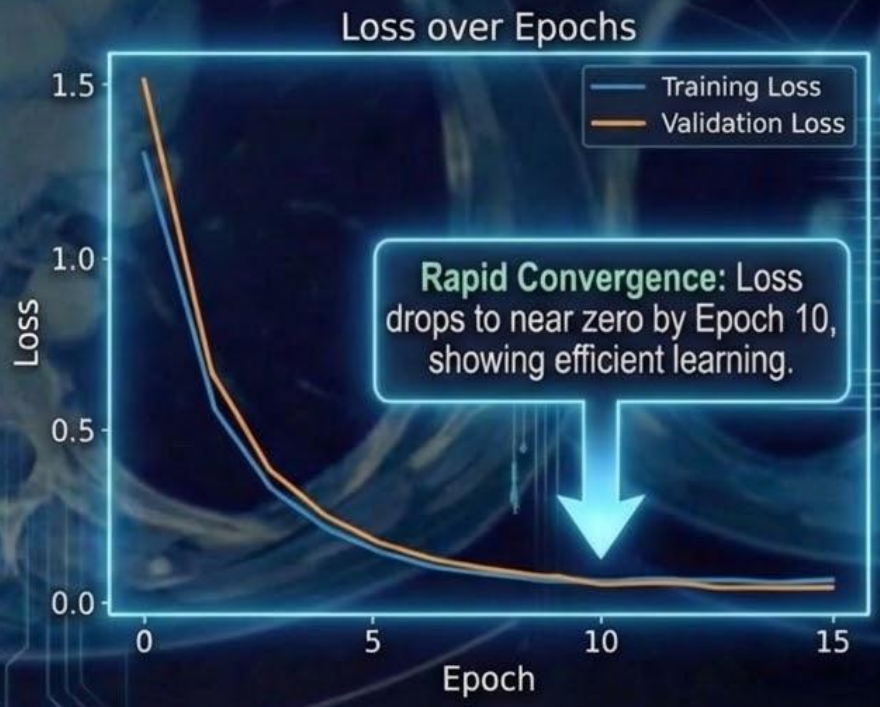
Malignant

Normal

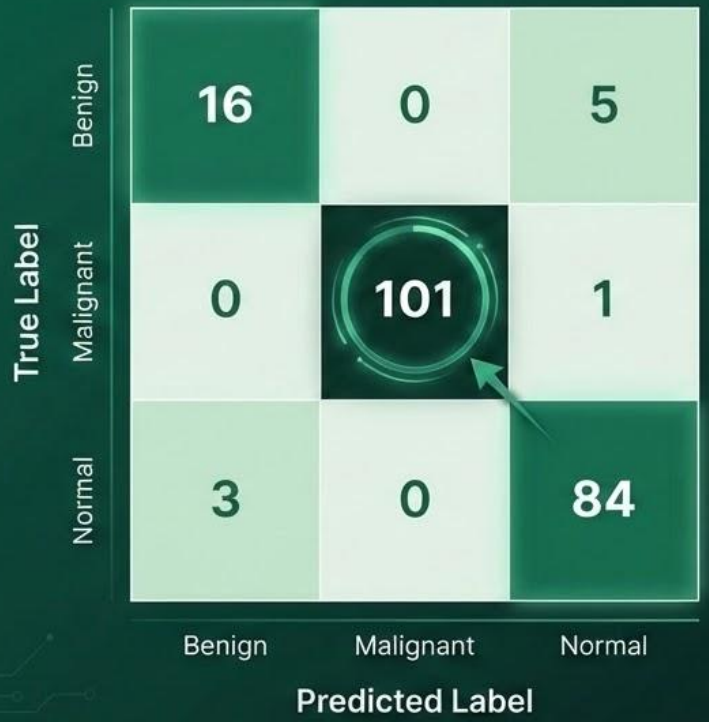
The Training Journey

Learning Curves & Stability





95.7% Real-World Accuracy



The Big Win

- **Highlight: 101**/102 Malignant cases caught.

In medicine, "Sensitivity" (catching the cancer) cancers) is the most vital metric. Our model streats exceptional sensitivity, ensuring almost all malignant cases are correctly identified.

Future Scope

Summary:

~~Flawed 100%~~
~~(Leaked Data)~~

→

Robust 95.7%
(Clean Data)

Diagnostic Tool

Next Steps:

Deploying as a secure cloud API for clinics.

Using **Grad-CAM** to show doctors exactly why the AI flagged a specific area.

Project Reflections & Closing

- **Scientific Integrity First:** The discovery of 43 duplicate images was a turning point; it proved that a "perfect" 100% accuracy often hides Data Leakage.
- **Architectural Shift:** Moving from a simple CNN (~45% accuracy) to VGG16 Transfer Learning demonstrated the necessity of "Specialist" models for complex medical textures.
- **Clinical Success:** Achieving 99% Sensitivity (101/102 malignant cases caught) ensures this tool prioritizes patient safety above all else.
- **Future Impact:** This project establishes a robust, clean-data baseline for deploying cloud-based diagnostic APIs and Grad-CAM explainability for radiologists.

Thank you for your time. I am now open to questions.

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